



education

Department:
Education
North West Provincial Government
REPUBLIC OF SOUTH AFRICA

PROVINCIAL ASSESSMENT

GRADE 11

MATHEMATICAL LITERACY P2

JUNE 2024

MARKS: 75

TIME: 1½ hours



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INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Start EACH question on a NEW page.
3. Number the answers correctly according to the numbering system used in this question paper.
4. You may use an approved calculator (non-programmable and non-graphical), unless stated otherwise.
5. Show ALL calculations clearly.
6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.
7. Indicate units of measurement, where applicable.
8. Maps and diagrams are NOT drawn to scale, unless stated otherwise.
9. Write neatly and legibly.

QUESTION 1

1.1

Neo is trying a recipe for making Greek stuffed eggplant. The recipe below shows the ingredients needed for making Greek stuffed eggplant.

PICTURE OF GREEK EGGPLANT	INGREDIENTS
	 2 x large eggplants  12 ounces (350 g) ground beef  100 g feta cheese  $\frac{1}{4}$ cup (60 ml) light cream  1 tablespoon of dried oregano  $\frac{1}{4}$ cup (60 ml) olive oil
Serves 	Preparation time: 10 min Cooking time : 40 min

Use the information above to answer the questions that follow.

1.1.1 Write down the maximum number of people that this recipe will cater for. (2)

1.1.2 State the recommended time it will take Neo to prepare and cook this dish. (2)

1.1.3 A $\frac{1}{4}$ cup of oil is equivalent to 60 ml. Determine the number of millilitres in one cup of oil. (2)

1.1.4 Neo will start to serve this dish at 6:37 pm.

Write down this time using the 24-hour clock format. (2)

1.2

The scale used on the seating plan of an examination room is 1:50.

1.2.1 Explain the meaning of the scale of the seating plan. (2)

1.2.2 Name ONE other type of scale that you have studied. (2)

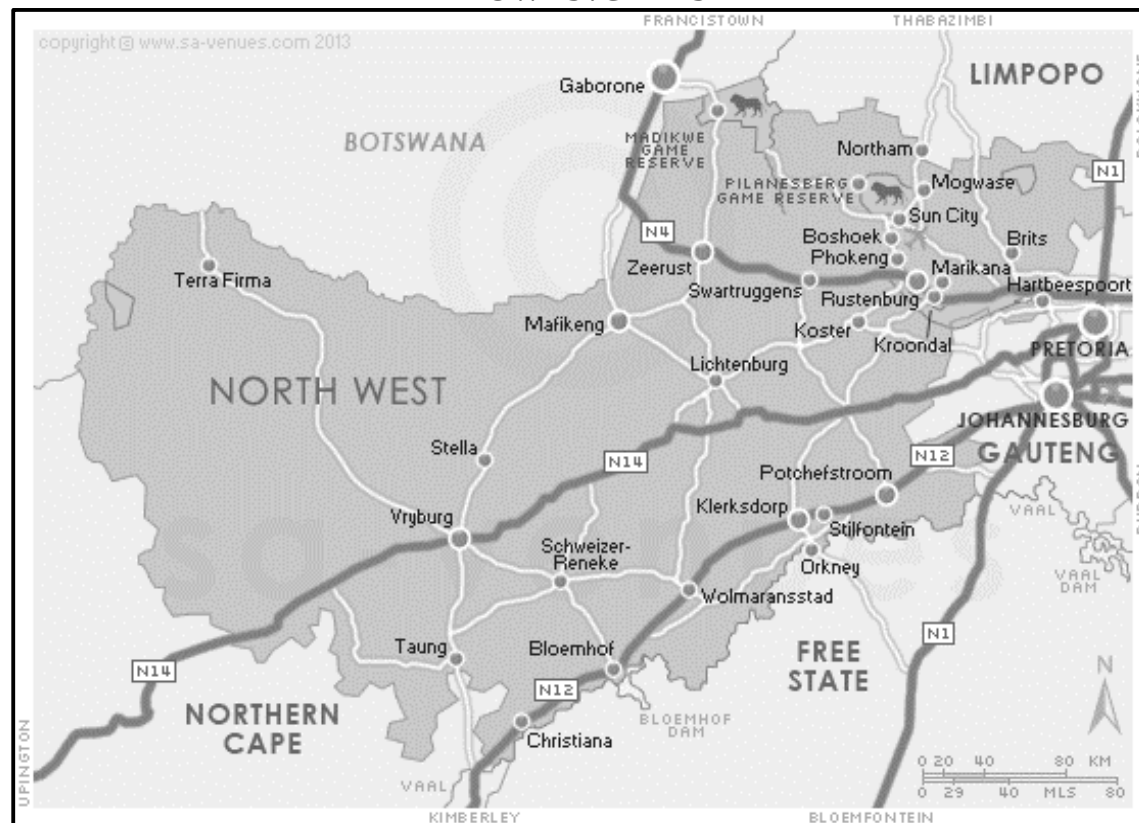
1.3 Define the concept *outcome* as used in probability. (2)
[14]

QUESTION 2

2.1

Below is a map of the North West province showing town/cities and roads.

MAP OF NORTH WEST SHOWING NATIONAL ROADS LEADING TO TOWNS /CITIES



Use the map above to answer the questions that follow.

2.1.1 Determine the total number of provinces other than North West shown on this map. (2)

2.1.2 The N4 measures 718 km, starts on the Botswana border and ends in Komatipoort on the Mozambique border.

Identify any TWO towns shown on the map where the N4 passes through. (2)

2.1.3 The area of the North West province is $104\,882\text{ km}^2$. The area of the whole of South Africa is $1\,220\,813\text{ km}^2$.

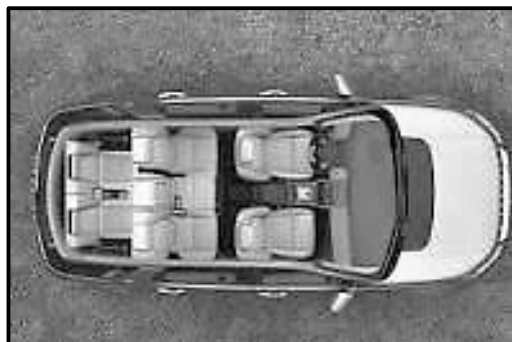
(a) Determine the percentage that the North West province make out of the whole of South Africa. (3)

(b) Give a reason why a traveller from Botswana needs a passport to enter the North West province. (2)

2.2

Mpho, Tshepo and two friends plan to drive to the Pilanesburg Game Reserve near Rustenburg. They will travel from Taung using a Land Rover. The vehicle has a fuel consumption of 6,42 liters/100 km. Its seating layout is shown below.

THE SEATING LAYOUT OF A LAND ROVER



Use the map and information above to answer the questions that follow.

2.2.1 State the view represented in the layout picture above. (2)

2.2.2 The distance from Taung to the Pilanesburg Game Reserve is 460 km. The Land Rover covers this distance in 4,5 hours.

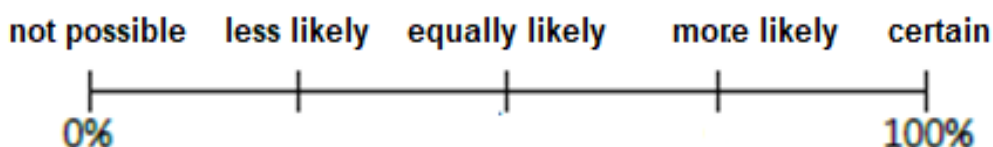
Calculate the average speed of this vehicle.

You may use the formula: **Distance = speed × time** (3)

2.2.3 Calculate the amount of petrol needed for the return trip. (4)

2.2.4 Tumi indicated to his friends that he was 25% sure that he will join them on this trip.

The probability scale below shows the likelihood for various probabilities.



Write down the likelihood that best describes Tumi's chances of going on this trip. (2)
[20]

QUESTION 3

3.1

Kea and Tumi are planning their 30th year anniversary this year. Kea wants a four-layered cylindrical vanilla and chocolate cake.

The diagram of the cake is shown below.

KEA'S FOUR LAYERED CYLINDRICAL VANILLA AND CHOCOLATE CAKE

Use the information above to answer the questions that follow.

3.1.1 Determine, in millimetres the total height of the cake. (3)

3.1.2 The cake was baked using 3 and a half pounds of butter.

Determine the mass of butter, in kilograms.

NOTE: 1 pound = 453,592 grams (3)

3.1.3 The bottom layer of the cake has a radius of 14 cm.

(a) Determine, in cm, the diameter of the bottom layer. (2)

(b) Verify, showing ALL the calculations, that the volume of the bottom layer of this cake is $9\,237,48\text{ cm}^3$.

You may use the formula:

Volume of a cylinder = $3,142 \times (\text{radius})^2 \times \text{height}$. (3)

3.1.4 The cake must be stored in the fridge at 5°C .
Convert this temperature to $^\circ\text{F}$.

You may use the formula: $^\circ\text{F} - 32^\circ = \left(\frac{9}{5} \times ^\circ\text{C}\right)$ (3)

3.2

Mrs Nku has two children, Tom and Neo. They attend two different schools. The information below describes Mrs Nku's routine on a particular morning.

- She drives the children to their respective schools.
- First she drops off Tom at point A.
- Then she takes Neo to her school at point B.
- Thereafter she returns home.

Mrs Nku's journey is illustrated in the graph below

Use the graph and information above to answer the questions that follow.

- 3.2.1 It took Mrs Nku 15 minutes to drive the 10 km from home to Tom's school.

Express 15 minutes as a fraction of an hour in decimal form. (2)

- 3.2.2 Indicate whether the following statements are TRUE or FALSE. If FALSE, correct the statement.

(a) It took Mrs Nku, 25 minutes to drive from Neo's school to home. (2)

(b) The distance from Tom's school to Neo's school is 15 km. (2)

- 3.2.3 Mrs Nku left her home at 6:47 in the morning to drive her kids to school. On her way back home, she stopped for 20 minutes at the shop to buy bread.

Verify, showing ALL calculations, whether she will arrive at her house before 8:00 a.m. (4)
[24]

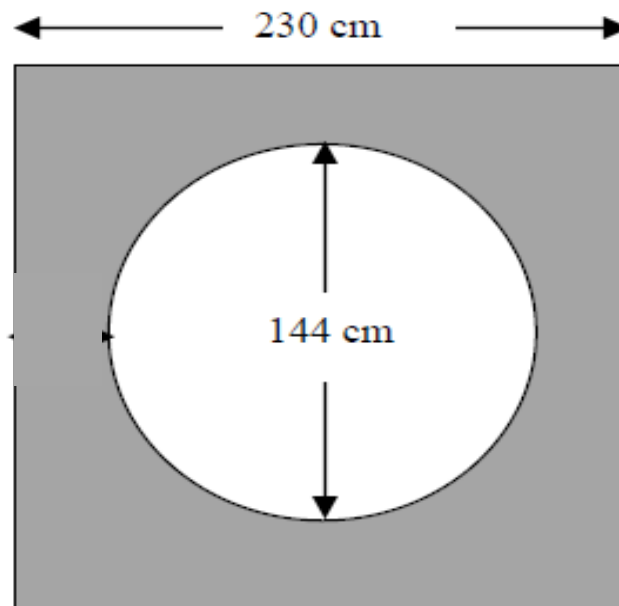
QUESTION 4

4.1

Sol installed a circular window in the centre of a square wall, as shown in the diagram below. He intends to paint the wall with 3 coats of paint.

The diameter of the circular window is 144 cm.

The length of each side of the square wall is 230 cm.



Use the information above to answer the questions that follow.

4.1.1 Calculate the circumference of the window.

You may use the formula: **Circumference of the circle** = $3,142 \times d$. (2)

4.1.2 Calculate, in cm^2 , the area of the wall that needs to be painted.

You may use the formulae:

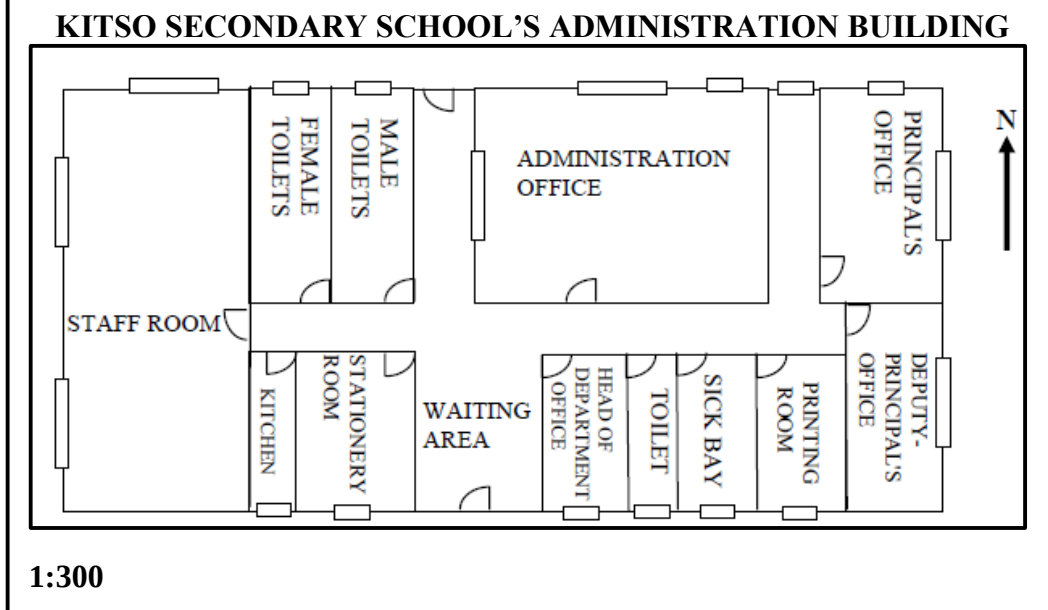
Area of a circle = $3,142 \times \left(\frac{d}{2}\right)^2$ where d is the diameter

Area of a square = s^2 , where s is the length of each side (6)

4.1.3 Convert the answer in QUESTION 4.1.2 to m^2 . (2)

4.2

The diagram below shows a scaled diagram of Kitso Secondary School's administration building.



Use the diagram above to answer the questions that follow.

4.2.1 Determine the number of offices in this administration building. (2)

4.2.2 The length of the printing room on the scale diagram is 2,2 cm.

Use the given scale above to calculate in metres, the actual length of the printing room. (3)

4.2.3 Give a reason why the administration building has a waiting area. (2)
(17)

TOTAL: 75